

WORKPLACE SAFETY MANUAL ? CHEMICAL HANDLING

1. INTRODUCTION

This manual provides comprehensive guidelines for the safe handling, storage, and disposal of hazardous chemicals used in the manufacturing process. All employees who work with or near chemicals must read this manual and complete the associated training program before commencing work.

2. CHEMICAL CLASSIFICATION

Chemicals are classified according to the Globally Harmonized System (GHS). Categories include: flammable liquids, corrosive substances, oxidizing agents, acute toxins, and environmental hazards. Each chemical container must display the appropriate GHS pictogram, signal word, and hazard statement.

3. PERSONAL PROTECTIVE EQUIPMENT

Required PPE for chemical handling includes: chemical-resistant gloves (nitrile or neoprene), safety goggles with splash guards, lab coat or chemical-resistant apron, closed-toe shoes, and a respirator when working with volatile compounds. PPE must be inspected before each use and replaced if damaged.

4. SPILL RESPONSE PROCEDURES

4.1 Minor Spills (less than 500 mL)

For minor spills of non-volatile, non-toxic chemicals: alert nearby workers, don appropriate PPE, contain the spill with absorbent materials, collect contaminated materials in a labeled waste bag, and clean the area with appropriate neutralizing solution. Report the incident to the shift supervisor.

4.2 Major Spills (500 mL or greater)

For major spills or spills involving highly hazardous chemicals: evacuate the immediate area, activate the chemical spill alarm, contact the Emergency Response Team (ERT) at extension 5555, do NOT attempt cleanup without ERT guidance, secure ventilation by opening fume hoods and exterior doors. The ERT leader will determine if external emergency services are required.

5. CHEMICAL STORAGE

Store chemicals in designated cabinets according to compatibility groups. Never store acids with bases. Flammable solvents must be kept in approved flammable storage cabinets. Maximum container size for bench storage is 2.5 L. All storage areas must have secondary containment.